

Ethan Nadler | Curriculum Vitae

University of California, San Diego
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Research

Galaxy Formation

- Understanding how the faintest galaxies formed and evolved;
- Modeling the connection between dwarf galaxies and dark matter halos.

Dark Matter

- Searching for signals of dark matter microphysics in small-scale structure data;
- Simulating structure formation beyond CDM using high-resolution simulations.

Near-field Cosmology

- Reconstructing primordial density fluctuations from local dwarf galaxy surveys;
- Combining dwarf galaxies and strong lensing to search for galaxy-free dark halos.

Positions

University of California, San Diego <i>Assistant Professor, Department of Astronomy & Astrophysics</i>	2025–
Carnegie Observatories & University of Southern California <i>Joint Postdoctoral Research Fellow, CTAC & USC Department of Physics and Astronomy</i>	2021–24

Education

Stanford University <i>Ph.D., Physics (Advisor: Risa H. Wechsler)</i> Thesis: Faint Galaxies and Small Halos: Probes of Galaxy Formation and Dark Matter	2021
University of California, Santa Barbara <i>B.S., Physics (Advisor: S. Peng Oh)</i> Thesis: Universality in the Structure and Abundance of Dark Matter Halos	2016

Grants

NSF Astronomy & Astrophysics Research Grant (PI; \$419,979) <i>Collaborative Research: Reconstructing Primordial Density Fluctuations using Near-Field Cosmology</i>	2024–
UC San Diego School of Physical Sciences Outreach Grant (PI; \$2,500) <i>The Preuss School + UCSD Astronomy & Astrophysics: Dark Matter & Scientific Programming</i>	2024–25
Carnegie Outreach Grant (PI; \$1,500) <i>CreateNow + Carnegie: Dark Matter & Data Visualization</i>	2022–24

Scientific Collaborations

Rubin LSST Dark Energy Science Collaboration: Co-Convener, Dark Matter Working Group	2025–
Satellites Around Galactic Analogs Survey: Member	2019–
DECam Local Volume Exploration Survey: Member	2019–
Rubin LSST Dark Energy Science Collaboration: Member	2018–
Dark Energy Survey: Member, Milky Way Working Group	2018–

Fellowships & Awards

Lattimer Research Fellow for Cross-disciplinary Explorations: UC San Diego	2026–
LSST Scialog Fellow: Research Corporation for Science Advancement	2025–
ACCESS Allocation: Optimizing Next-generation Cosmological Zoom-in Simulations	2025–
Faculty Fellow: San Diego Supercomputer Center	2025
XSEDE Allocation: Cosmological Simulations of Milky Way Analogs with Galactic Disks	2022–23
XSEDE Allocation: Simulations of Milky Way Halos with Large Magellanic Cloud Analogs	2020–22
NSF Graduate Research Fellow: National Science Foundation	2018–21
Faculty Committee Commendation of Excellence: UCSB College of Creative Studies	2016
Outstanding Senior Award: UCSB Department of Physics	2016

Teaching

Professor (UCSD)	2025–
○ <i>ASTR 2: Galaxies and the Universe</i> (Spring '25, '26);	
○ <i>ASTR 122: Physical Cosmology</i> (Winter '26).	
Guest Lecturer (USC)	2022
○ <i>Advanced Cosmology:</i> Lecture on <i>Structure Formation & Galaxies</i> .	
Textbook Co-Author (UC Davis)	2022
○ <i>A Cosmology Workbook:</i> 31: Structure Formation , 32: Galaxy Formation .	
Teaching Assistant (Stanford)	2017–21
○ <i>Structure Formation & Galaxy Formation, Modern Astrophysics, Cosmology & Extragalactic Astrophysics, Origin & Development of the Cosmos, Electricity & Magnetism.</i>	

Service

Workshop Organizer (Spec-S5 Dark Matter Meeting I, II , KICP/NOIRLab)	2025–
Conference Organizer (GalFRESKA , UCSD)	2025
ArXiv Hour Committee Chair (UCSD Astronomy & Astrophysics)	2025–
Postdoctoral Scholar Mentor (UCSD Astronomy & Astrophysics)	2025–
Graduate Advisory Committee Member (UCSD Astronomy & Astrophysics)	2025–
Graduate Admissions Committee Member (UCSD Astronomy & Astrophysics)	2024–25
Conference Coordinator (Cosmic Signals of Dark Matter Physics: New Synergies , KITP)	2024
Proposal Review Panel Member (NASA 2022; NSF 2025, 2026; STScI 2026)	2022–
USC Physics and Astronomy Climate Committee (Postdoctoral & Staff Representative)	2021–24
Journal Referee (<i>ApJ</i> , <i>ApJL</i> , <i>Astronomy & Astrophysics</i> , <i>Astroparticle Physics</i> , <i>JCAP</i> , <i>MNRAS</i> , <i>Nature</i> , <i>PRD</i> , <i>PRL</i> , <i>Reviews of Modern Physics</i> , <i>Language & Cognition</i> , <i>Lingua</i>)	2019–

Outreach

Osher Lifelong Learning Institute Astronomy & Astrophysics Master Class (Speaker)	2026
Lyncean Group of San Diego Lecture (Speaker)	2025
UCSD STARTastro: Astronomy Workshop (Speaker)	2025
UCSD Stellar Beginnings: Astronomy & Astrophysics Department Launch (Speaker)	2025
UCSD Preuss High School: Dark Matter & Scientific Programming (Course Instructor)	2025
Carnegie Observatories Lectures at the Huntington (Speaker)	2023
USC Hybrid High School: Dark Matter & Data Visualization (Course Instructor)	2022
UCSB Physics NSF REU (Speaker)	2021–25

Mentoring

Postdoc Advisor (bold: current)

2025–

- **Sandip Roy**, UCSD Schmidt AI Fellow: Emulating small-scale dark matter structure

Graduate Student Advisor (bold: current; †: thesis student)

2025–

- †**Ollie Jackson**, UCSD: Strong lensing dark matter substructure modeling and constraints
- †**Wisha Wanichwecharunguang**, UCSD: Near-field cosmology using semi-analytic models
- †**Zewei Wu**, UCSD: Hydrodynamic simulations of ultra-faint dwarf satellite galaxies

Undergraduate Student Advisor (bold: current; †: thesis student)

2025–

- **Stephen Canino**, UCSD '28: Subhalo abundance evolution in zoom-in simulations
- **Kimmy Dang**, Yale '28: Large Magellanic Cloud satellite populations across environments
- Steve Du, UCSD '25: Sterile neutrino dark matter constraints from small-scale structure
- **Bocheng Feng**, Peking '26: Dark matter halo pseudo phase space density profiles (see [B. Feng, E. O. Nadler et al. 2026](#))
- **Roxanne Lai**, UCSD '28: Subhalo abundance evolution in zoom-in simulations
- †**Sophia Um**, UCSD, '26: Environmental dependence of dwarf galaxy star formation

Graduate Student Co-advisor (bold: current)

2021–

- Niussha Ahvazi, UCR '24 → UVA: Semi-analytic modeling of dwarf galaxy formation and evolution (see [N. Ahvazi, A. Benson, L. Sales, E. O. Nadler et al. 2024](#));
- **Wendy Crumrine**, USC: Constraining dark matter–radiation interactions with small-scale structure (see [W. Crumrine, E. O. Nadler et al. 2024](#); [W. Crumrine, D. Pang, E. O. Nadler et al. 2026](#));
- **Tara Dacunha**, Stanford: Satellite galaxy disruption and stellar stream modeling;
- Elise Darragh-Ford, Stanford '24: Searching for dwarf galaxies and stellar streams in *Gaia* data (see [E. Darragh-Ford, E. O. Nadler et al. 2021](#));
- Trey Driskell, USC '25: Semi-analytic modeling of structure and galaxy formation (see [T. Driskell, E. O. Nadler et al. 2022](#));
- Noah Glennon, UNH '23: Soliton orbital evolution in self-interacting axion dark matter (see [N. Glennon, E. O. Nadler et al. 2022](#); [N. Glennon, N. Musoke, E. O. Nadler et al. 2024](#));
- **Demao Kong**, UCR: Modeling SIDM subhalo populations in the Milky Way and strong lens analogs (see [D. Kong, H.-B. Yu, E. O. Nadler et al. 2025](#); [D. Kong, E. O. Nadler, H.-B. Yu 2025](#));
- Sidney Mau, Stanford '25: Constraining the dark matter particle lifetime with dwarf galaxies (see [S. Mau, E. O. Nadler et al. 2022](#));
- **Siddhesh Raut**, USC: Self-interacting dark matter halo gravothermal evolution modeling; (see [S. Raut, E. O. Nadler, and A. Benson 2026](#));
- Yunchong Wang, Stanford '24: Empirically modeling dwarf galaxy star formation histories (see [Y. Wang, E. O. Nadler et al. 2021, 2024a](#); [Y. Wang, P. Mansfield, E. O. Nadler et al. 2024b](#));
- **James Wen**, USC: Hydrodynamic simulations of dark matter–baryon interactions;
- Xingyu Zhang, Tsinghua '25: Modeling stellar stream perturbers as SIDM subhalos (see [X. Zhang, H.-B. Yu, D. Yang, and E. O. Nadler 2025](#)).

Undergraduate Student Co-advisor

2018–

- Deveshi Buch, Stanford '24: Cosmological zoom-in simulations of Milky Way-like systems (see [D. Buch, E. O. Nadler et al. 2024](#));
- Shuxing Fang, USC '22: Large Magellanic Cloud infall in self-interacting dark matter;
- Abigail Lee, UPenn '19 → UChicago: Subhalo disruption in galaxy clusters.
- Nyal McCrea (Simons–NSBP Scholar), CWU '22: Zoom-in simulation visualizations;
- Ellen Min, Caltech '24: Code development and Python implementation for [Galacticus](#);
- Nicel Mohamed-Hinds, Stanford '19 → UW: Emulating hydrodynamic zoom-in simulations;
- Ezra Msolla (Simons–NSBP Scholar), UToronto '25: Neutrino self-interaction impact on cosmic structure
- Veronica Pratt, Stanford '23 → Tufts: Statistics of Large Magellanic Cloud analogs in the SAGA Survey;
- Juan Quiroz, Caltech '24: Modeling subhalo evolution in decaying dark matter models;
- Resherle Verna, USC '20 → UT Austin: Subhalo populations in SIDM + hydrodynamic simulations;
- Logan White (Simons–NSBP Scholar), NCSU '25: Halo mass function evolution beyond CDM;

Media

Space.com , <i>'Dark subhaloes' may explain why galaxies seem to form pre-determined shapes</i>	2026
Astronomy Magazine , <i>How Weird is the Milky Way?</i>	2025
USC Today , <i>Scientists code Milky Way twin galaxies to better understand dark matter</i>	2025
UC San Diego Today , <i>A New Astronomy Programs Helps Young Minds Tackle Big Questions</i>	2025
UC San Diego Today , <i>Do "Completely Dark" Dark Matter Halos Exist?</i>	2025
UC Riverside News , <i>New dark matter theory explains two puzzles in astrophysics</i>	2023
Quanta Magazine , <i>In a Monster Star's Light, a Hint of Darkness</i>	2023
KIPAC Research Highlight , <i>Between the worlds of the visible and invisible lies: Dark Matter</i>	2021
Fermilab Press Release , <i>DES census of the smallest galaxies hones the search for dark matter</i>	2020
SLAC Press Release , <i>Milky Way satellites reveal link between dark matter and galaxy formation</i>	2020
AAS Nova Research Highlight , <i>Constraining collisions of dark matter</i>	2019
SLAC Press Release , <i>Satellite galaxies provide new clues about dark matter</i>	2019
KIPAC Research Highlight , <i>Dark matter subhalo disruption: insights from machine learning</i>	2018

Presentations

<i>Review: Unlocking Exotic Dark Matter with the Low Surface Brightness Universe</i> EPFL, European Astronomical Society Annual Meeting	2026
<i>Is Dark Matter Falsifiable? Lessons from Small-scale Cosmic Structure</i> UC Irvine, Foundations of Physics 2026 [abstract]	2026
<i>Dark Matter Insights from Small-scale Structure</i> CERN, Beyond the Standard Model Forum Stony Brook/Brookhaven, YITP Seminar	2025–
<i>Which Dark Matter Halos Form Stars?</i> UCSC 2025 Galaxy Workshop [video]	2025–
<i>Review: Satellite Galaxies and Stellar Streams as Probes of SIDM</i> Valencia Instituto de Física Corpuscular, Small-scale Structure of the Universe & SIDM [slides]	2025
<i>Revealing Dark Matter and Galaxy Formation with Small-Scale Structure</i> UCLA, Astronomy Colloquium [video] University of Washington, Astronomy Colloquium Stanford/KIPAC, Astrophysics Colloquium [video] UCSC, Astronomy & Astrophysics Colloquium UC Merced, Physics Colloquium Carnegie Observatories, Colloquium Rice, Astronomy & Astrophysics Seminar UC San Diego, Astrophysics Colloquium Caltech, TAPIR Seminar	2024–
<i>COZMIC: Cosmological Zoom-in Simulations with Initial Conditions Beyond CDM</i> UCLA, Dark Matter 2025 [slides] Princeton, Dark Cosmos Seminar PACIFIC Conference	2024–25
<i>Review: What can Dwarf Galaxies Reveal about the Nature of Dark Matter?</i> Dynamical Tracers of the Nature of Dark Matter [slides] Durham, Small Galaxies, Cosmic Questions - II [slides]	2024–25

<i>Dark Matter Physics in the Sky</i> KITP Blackboard Talk [video]	2024
<i>Forecasts for Galaxy Formation and Dark Matter Constraints from Dwarf Galaxy Surveys</i> LBNL, Fundamental Physics from Future Spectroscopic Surveys [slides]	2024
<i>SIDM (Sub)halos in Milky Way and Strong Lens Analogs</i> Pollica Physics Centre, Self-Interacting Dark Matter Models, Simulations and Signals [slides]	2023
<i>Cosmological Simulations with Novel Dark Matter Physics</i> UCLA, Dark Matter 2023 [slides] KICP, Astronomy & Astrophysics Seminar	2023
<i>The Faint End of the Galaxy–Halo Connection</i> KITP, Building a Physical Understanding of Galaxy Evolution [video]	2023
<i>Dark Matter Constraints from Small-Scale Structure</i> CERN, New Physics from Galaxy Clustering [video , slides]	2022
<i>Symphony: Cosmological Zoom-in Simulations over Four Decades of Host Halo Mass</i> Caltech, GalFRESKA [slides]	2022
<i>Dark Matter Physics + Rubin LSST</i> LSST DESC, CosmoPalooza [slides]	2022
<i>Towards Precision Near-Field Cosmology</i> KIPMU, Astro Lunch Seminar UC Riverside, Astronomy Seminar Fermilab, Cosmic Physics Center Seminar [video]	2021–22
<i>Dark Matter Constraints from a Unified Analysis of Strong Lenses and Satellite Galaxies</i> LSST DESC, Dark Matter Working Group Virginia Tech Center for Neutrino Physics, Journal Club	2021
<i>The Faintest Galaxies and their Dark Matter Halos</i> Caltech, TAPIR Seminar Harvard-Smithsonian Center for Astrophysics, GCSP Seminar [video] International Centre for Theoretical Sciences, Less Travelled Path of Dark Matter [video , slides] UC Berkeley Center for Cosmological Physics, Cosmology Seminar [slides] STScI, The Local Group: Assembly and Evolution KITP, The Galaxy–Halo Connection Across Cosmic Time: Recent Updates [video] LineA, Webinar [video , slides] KIPAC, Astrophysics Colloquium [video] Fermilab, New Perspectives [slides] BSM Pandemic Seminar [video , slides]	2020–21
<i>Milky Way Satellites: Probes of Dark Matter Microphysics</i> University of Chicago, Cosmic Controversies [slides] KICP, LSST Dark Matter Workshop [slides] Institute for Advanced Study, Astro Coffee Johns Hopkins, High Energy Physics/Cosmology Seminar UC Berkeley, LSST DESC Winter Collaboration Meeting	2019
<i>Modeling Subhalos and Satellites in Milky Way-like Systems</i> KICP, Near-Field Cosmology with DES DR1 [slides]	2018

First-authored Publications (Google Scholar h -index: 37; *: ≥ 100 Google Scholar citations)

21. **E. O. Nadler**, M. Amin, R. H. Wechsler, M. Sten Delos, A. Benson, and V. Gluscevic. *Warm, not Fuzzy: Generalized Ultralight Dark Matter Limits from Milky Way Satellites*. [2605.15371](#) (ApJL submitted).
20. **E. O. Nadler**, V. Gluscevic, and A. Benson. *The Effects of Linear Matter Power Spectrum Enhancement on Dark Matter Substructure*. 2025, [ApJ](#), **993**, 17.
19. **E. O. Nadler**, D. Kong, D. Yang, and H.-B. Yu. *SIDM Concerto: Compilation and Data Release of Self-interacting Dark Matter Zoom-in Simulations*. 2025, [ApJ](#), **991**, 69.
18. **E. O. Nadler**, R. An, D. Yang, H.-B. Yu, A. Benson, and V. Gluscevic. *COZMIC. III. Cosmological Zoom-in Simulations of Self-interacting Dark Matter with Suppressed Initial Conditions*. 2025, [ApJ](#), **986**, 129.
17. **E. O. Nadler**, R. An, V. Gluscevic, A. Benson, and X. Du. *COZMIC. I. Cosmological Zoom-in Simulations with Initial Conditions Beyond Cold Dark Matter*. 2025, [ApJ](#), **986**, 127.
16. **E. O. Nadler** & A. Benson. *Semianalytic model for decaying dark matter halos*. 2025, [PRD](#), **111**, 103522.
15. **E. O. Nadler**. *The Impact of Molecular Hydrogen Cooling on the Galaxy Formation Threshold*. 2025, [ApJL](#), **983**, L23.
14. **E. O. Nadler**, V. Gluscevic, T. Driskell, R. H. Wechsler, L. A. Moustakas, *et al.* *Forecasts for Galaxy Formation and Dark Matter Constraints from Dwarf Galaxy Surveys*. 2024, [ApJ](#), **967**, 61.
13. **E. O. Nadler**, D. Yang, and H.-B. Yu. *A Self-interacting Dark Matter Solution to the Extreme Diversity of Low-mass Halo Properties*. 2023, [ApJL](#), **958**, L39.
12. **E. O. Nadler**, P. Mansfield, Y. Wang, X. Du *et al.* *Symphony: Cosmological Zoom-in Simulation Suites over Four Decades of Host Halo Mass*. 2023, [ApJ](#), **945**, 159.
11. **E. O. Nadler**, A. Benson, T. Driskell, X. Du, and V. Gluscevic. *Growing the first galaxies' merger trees*. 2023, [MNRAS](#), **521**, 3201.
10. **E. O. Nadler**, A. Banerjee, S. Adhikari, Y.-Y. Mao, and R. H. Wechsler. *The Effects of Dark Matter and Baryonic Physics on the Milky Way Subhalo Population in the Presence of the Large Magellanic Cloud*. 2021, [ApJL](#), **920**, L11.
- *9. **E. O. Nadler**, S. Birrer, D. Gilman, R. H. Wechsler, X. Du, A. Benson, A. Nierenberg, and T. Treu. *Dark Matter Constraints from a Unified Analysis of Strong Gravitational Lenses and Milky Way Satellite Galaxies*. 2021, [ApJ](#), **917**, 7.
- *8. **E. O. Nadler** & A. Drlica-Wagner *et al.* (DES Collaboration). *Constraints on Dark Matter Properties from Observations of Milky Way Satellite Galaxies*. 2021, [PRL](#), **126**, 091101.
7. **E. O. Nadler**, A. Banerjee, S. Adhikari, Y.-Y. Mao, and R. H. Wechsler. *Signatures of Velocity-dependent Dark Matter Self-interactions in Milky Way-mass Halos*. 2020, [ApJ](#), **896**, 112.
- *6. **E. O. Nadler** & R. H. Wechsler *et al.* (DES Collaboration). *Milky Way Satellite Census. II. Galaxy-Halo Connection Constraints Including the Impact of the Large Magellanic Cloud*. 2020, [ApJ](#), **893**, 48.
- *5. **E. O. Nadler**, V. Gluscevic, K. K. Boddy, and R. H. Wechsler. *Constraints on Dark Matter Microphysics from the Milky Way Satellite Population*. 2019, [ApJL](#), **878**, L32.

- *4. **E. O. Nadler**, Y.-Y. Mao, G. M. Green, and R. H. Wechsler. *Modeling the Connection between Subhalos and Satellites in Milky Way-like Systems*. 2019, [ApJ](#), **873**, 34.
3. **E. O. Nadler**, Y.-Y. Mao, R. H. Wechsler, S. Garrison-Kimmel, and A. Wetzel. *Modeling the Impact of Baryons on Subhalo Populations with Machine Learning*. 2018, [ApJ](#), **859**, 129.
2. **E. O. Nadler**, A. Perko, and L. Senatore. *On the bispectra of very massive tracers in the Effective Field Theory of Large-Scale Structure*. 2018, [JCAP](#), **1**, 058.
1. **E. O. Nadler**, S. P. Oh, and S. Ji. *On the apparent power law in CDM halo pseudo-phase space density profiles*. 2017, [MNRAS](#), **470**, 500.

Co-authored Publications (*: ≥ 100 Google Scholar citations)

57. A. Benson, **E. O. Nadler**, X. Du, and V. Gluscevic. *A Unified Halo Mass Function Across Dark Matter Models from High-Resolution Multi-Scale Simulations*. [2606.12137](#) (OJAp submitted).
 ○ Major contributions: Conceptualized semi-analytic model calibration on zoom-in simulations.
56. W. Crumrine, D. Pang, **E. O. Nadler**, A. Benson, and V. Gluscevic. *Mixed Dark Matter: Limits from the Milky Way Satellite Galaxies*. [2606.10006](#) (PRD submitted).
 ○ Major contributions: Interpreted fractional interacting and fuzzy dark matter constraints and forecasts.
55. S. Raut, **E. O. Nadler**, and A. Benson. *An Extended Parametric Model for Self-interacting Dark Matter Halos*. [2604.22013](#) (ApJ submitted).
 ○ Major contributions: Provided SIDM simulations and interpreted effects of mass accretion on halo evolution.
54. A. Engelhardt, F. Munshi, A. H. G. Peter, **E. O. Nadler et al.** *MARVELously Dark: the gravothermal evolution of dwarf halos in velocity-dependent SIDM*. [2601.23264](#) (JCAP submitted).
53. E. Darragh-Ford & N. Garavito-Camargo *et al.* (incl. **E. O. Nadler**). *Shaping the Milky Way. II. The dark matter halo's response to the LMC's passage in a cosmological context*. [2511.02031](#) (ApJ submitted).
52. D. Kong, **E. O. Nadler**, and H.-B. Yu. *Strong Lensing Perturbors from the SIDM Concerto Suite*. [2510.01491](#) (PRD submitted).
 ○ Major contributions: Interpreted environmental dependence of SIDM predictions and comparisons to data.
51. J. Peñarrubia & **E. O. Nadler**. *A dynamical attractor in the evolution of dwarf spheroidal galaxies*. 2026, [MNRAS](#), **548**, [stag803](#).
 ○ Major contributions: Conceptualized application to Milky Way satellites and interpreted scaling relations.
50. C. Y. Tan, A. Drlica-Wagner, A. B. Pace, W. Cerny, **E. O. Nadler et al.** (DELVE Collaboration). *DELVE Milky Way Satellite Galaxy Census I: Satellite Population and Survey Selection Function*. 2026, [ApJ](#), **1000**, 87.
 ○ Major contributions: Interpreted empirical constraints on total Milky Way satellite population and its anisotropy.
49. B. Feng, **E. O. Nadler**, S. P. Oh, and S. Ji. *The non-universal pseudo-phase-space density profiles of symphony host haloes*. 2026, [MNRAS](#), **546**, [stag215](#).
 ○ Major contributions: Piloted correlations between PPSD and halo properties and analysis pipeline.
48. Y. Asali *et al.* (SAGA Survey, incl. **E. O. Nadler**). *The SAGA Survey. VI. The Size–Mass Relation for Low-mass Galaxies Across Environments*. 2025, [ApJ](#), **995**, 79.
47. E. Kado-Fong *et al.* (SAGA Survey, incl. **E. O. Nadler**). *SAGAbg. III. Environmental Stellar Mass Functions, Self-quenching, and the Stellar-to-halo Mass Relation in the Dwarf Galaxy Regime*. 2025, [ApJ](#), **994**, 231.
46. D. Kong, H.-B. Yu, **E. O. Nadler et al.** *Novel challenges in tracking self-interacting dark matter subhalos*. 2025, [JCAP](#), **10**, 074.
 ○ Major contributions: Conceptualized subhalo finder comparison and interpreted subhalo population results.

45. D. C. Baxter, A. L. Coil, **E. O. Nadler et al.** *Quantifying the Impact of Incompleteness on Identifying and Interpreting Galaxy Protocluster Populations with the TNG-Cluster Simulation*. 2025, [ApJ, 990, 225](#).
44. M. S. Fischer, K. Dolag, M. Garny, V. Gluscevic, F. Groth, and **E. O. Nadler**. *N-body simulations of dark matter-baryon interactions*. 2025, [A&A 700, A145](#).
43. K. Tsiane, S. Mau, A. Drlica-Wagner, J. L. Carlin, P. S. Ferguson, K. Bechtol, **E. O. Nadler et al.** (DESC Collaboration). *Predictions for the Detectability of Milky Way Satellite Galaxies and Outer-Halo Star Clusters with the Vera C. Rubin Observatory*. 2025, [OJAp, 8, 89](#).
- Major contributions: Co-developed predictions for Rubin Milky Way satellite luminosity function constraints.
42. Y. Wang, P. Mansfield, **E. O. Nadler**, E. Darragh-Ford, and R. H. Wechsler. *EDEN: Exploring Disks Embedded in N-body simulations of Milky-Way-mass halos from Symphony*. 2025, [ApJ, 986, 147](#).
- Major contributions: Co-developed disk potential simulation algorithm and interpreted disruption results.
41. R. An, **E. O. Nadler**, A. Benson, and V. Gluscevic. *COZMIC. II. Cosmological Zoom-in Simulations with Fractional non-CDM Initial Conditions*. 2025, [ApJ, 986, 128](#)
- Major contributions: Developed pipeline for fractional non-CDM simulations and derived constraints.
40. S. Ando, S. Horigome, **E. O. Nadler**, D. Yang, and H.-B. Yu. *SASHIMI-SIDM: semi-analytical subhalo modelling for self-interacting dark matter at sub-galactic scales*. 2025, [JCAP, 2, 053](#).
- Major contributions: Derived analytic prediction for core-collapsed fraction and interpreted subhalo results.
39. W. Crumrine, **E. O. Nadler**, R. An, and V. Gluscevic. *Dark matter coupled to radiation: Limits from the Milky Way satellites*. 2025, [PRD, 111, 023530](#).
- Major contributions: Co-developed and interpreted dark matter-radiation scattering constraints.
38. D. Yang, **E. O. Nadler**, and H.-B. Yu. *Testing the parametric model for self-interacting dark matter using matched halos in cosmological simulations*. 2025, [Physics of the Dark Universe, 47, 101807](#).
- Major contributions: Provided cosmological SIDM simulations and interpreted parametric model results.
37. X. Zhang, H.-B. Yu, D. Yang, and **E. O. Nadler**. *The GD-1 Stellar Stream Perturber as a Core-collapsed Self-interacting Dark Matter Halo*. 2025, [ApJL, 978, L23](#).
- Major contributions: Developed subhalo resimulation method and interpreted GD-1 results.
36. Y. Wang, **E. O. Nadler et al.** (SAGA Survey). *The SAGA Survey. V. Modeling Satellite Systems around Milky Way-mass Galaxies with Updated UNIVERSEMACHINE*. 2024, [ApJ, 976, 119](#).
- Major contributions: Interpreted galaxy-halo connection constraints; co-developed modeling pipeline.
35. M. Geha et al. (SAGA Survey, incl. **E. O. Nadler**). *The SAGA Survey. IV. The Star Formation Properties of 101 Satellite Systems around Milky Way-mass Galaxies*. 2024, [ApJ, 976, 118](#).
34. Y.-Y. Mao et al. (SAGA Survey, incl. **E. O. Nadler**). *The SAGA Survey. III. A Census of 101 Satellite Systems around Milky Way-mass Galaxies*. 2024, [ApJ, 976, 117](#).
33. E. Kado-Fong et al. (SAGA Survey, incl. **E. O. Nadler**). *SAGAbg II: The Low-mass Star-forming Sequence Evolves Significantly between $0.05 < z < 0.21$* . 2024, [ApJ, 976, 83](#).
32. D. Buch, **E. O. Nadler**, R. H. Wechsler, and Y.-Y. Mao. *Milky Way-est: Cosmological Zoom-in Simulations with Large Magellanic Cloud and Gaia-Sausage-Enceladus Analogs*. 2024, [ApJ, 971, 79](#).
- Major contributions: Piloted constrained zoom-in simulations and co-developed analysis pipeline.
31. P. Mansfield, E. Darragh-Ford, Y. Wang, **E. O. Nadler**, B. Diemer, and R. H. Wechsler. *SYMFIND: Addressing the Fragility of Subhalo Finders and Revealing the Durability of Subhalos*. 2024, [ApJ, 970, 178](#).
30. X. Du et al. (incl. **E. O. Nadler**). *Tidal evolution of cored and cuspy dark matter halos*. 2024, [PRD, 110, 023019](#).
29. E. Kado-Fong et al. (SAGA Survey, incl. **E. O. Nadler**). *SAGAbg. I. A Near-unity Mass-loading Factor in Low-mass Galaxies via Their Low-redshift Evolution in Stellar Mass, Oxygen Abundance, and Star*

Formation Rate. 2024, [ApJ](#), 966, 129.

28. [N. Ahvazi](#), A. Benson, L. V. Sales, **E. O. Nadler** *et al.* *A comprehensive model for the formation and evolution of the faintest Milky Way dwarf satellites*. 2024, [MNRAS](#), 529, 3387.

○ Major contributions: Interpreted galaxy–halo connection and Milky Way satellite predictions.

27. [N. Glennon](#), N. Musoke, **E. O. Nadler**, C. Prescod-Weinstein, and R. H. Wechsler. *Dynamical friction in self-interacting ultralight dark matter*. 2024, [PRD](#), 109, 063501.

26. D. Yang, **E. O. Nadler**, H.-B. Yu, and Y.-M. Zhong. *A parametric model for self-Interacting dark matter halos*. 2024, [JCAP](#), 2, 032.

○ Major contributions: Ran cosmological SIDM simulations and interpreted parametric model performance.

25. M. McNanna, K. Bechtol, S. Mau, **E. O. Nadler** *et al.* (DES Collaboration). *A Search for Faint Resolved Galaxies Beyond the Milky Way in DES Year 6: A New Faint, Diffuse Dwarf Satellite of NGC 55*. 2024, [ApJ](#), 961, 126

○ Major contributions: Developed dwarf galaxy population predictions and interpreted NGC 55 satellite.

24. P. Hopkins, **E. O. Nadler**, M. Grudić, X. Shen *et al.* *Novel conservative methods for adaptive force softening in collisionless and multispecies N-body simulations*. 2023, [MNRAS](#), 525, 5951.

○ Major contributions: Conceptualized softening algorithms and interpreted cosmological simulation results.

23. E. Darragh-Ford *et al.* (DESI Collaboration, incl. **E. O. Nadler**). *Target Selection and Sample Characterization for the DESI LOW-Z Secondary Target Program*. 2023, [ApJ](#), 954, 149.

22. R. An, V. Gluscevic, **E. O. Nadler**, and Y. Zhang. *Can Neutrino Self-interactions Save Sterile Neutrino Dark Matter?* 2023, [ApJL](#), 954, L18.

○ Major contributions: Developed sterile neutrino limits and interpreted production mechanism constraints.

21. A. Banerjee, S. Das, A. Maharana, **E. O. Nadler**, and R. K. Sharma. *Nonthermal warm dark matter limits from small-scale structure*. 2023, [PRD](#), 108, 043518.

○ Major contributions: Derived small-scale structure constraints and interpreted results.

20. W. Cerny *et al.* (DELVE Collaboration, incl. **E. O. Nadler**). *Six More Ultra-faint Milky Way Companions Discovered in the DECam Local Volume Exploration Survey*. 2023, [ApJ](#), 953, 1.

19. D. Yang, **E. O. Nadler**, and H.-B. Yu. *Strong Dark Matter Self-interactions Diversify Halo Populations within and surrounding the Milky Way*. 2023, [ApJ](#), 949, 67.

○ Major contributions: Performed cosmological SIDM simulations and interpreted dwarf galaxy predictions.

18. S. Yang, X. Du, Z. C Zeng, A. Benson, F. Jiang, **E. O. Nadler** *et al.* *Gravothermal Solutions of SIDM Halos: Mapping from Constant to Velocity-dependent Cross Section*. 2023, [ApJ](#), 946, 47.

17. S. Wagner-Carena, J. Aalbers, S. Birrer, **E. O. Nadler** *et al.* *From Images to Dark Matter: End-to-end Inference of Substructure From Hundreds of Strong Gravitational Lenses*. 2023, [ApJ](#), 942, 75.

16. [T. Driskell](#), **E. O. Nadler**, J. Mirocha, A. Benson, K. K. Boddy *et al.* *Structure formation and the global 21-cm signal in the presence of Coulomb-like dark matter-baryon interactions*. 2022, [PRD](#), 106, 103525.

○ Major contributions: Interpreted structure formation predictions for interacting dark matter models.

15. [N. Glennon](#), **E. O. Nadler**, N. Musoke, A. Banerjee, C. Prescod-Weinstein, and R. H. Wechsler. *Tidal disruption of solitons in self-interacting ultralight axion dark matter*. 2022, [PRD](#), 105, 123540.

○ Major contributions: Conceptualized and interpreted soliton tidal disruption simulations.

14. [S. Mau](#), **E. O. Nadler**, R. H. Wechsler, A. Drlica-Wagner, K. Bechtol *et al.* (DES Collaboration). *Milky Way Satellite Census. IV. Constraints on Decaying Dark Matter from Observations of Milky Way Satellite Galaxies*. 2022, [ApJ](#), 932, 128.

○ Major contributions: Performed cosmological decaying dark matter simulations and derived constraints.

13. J. Bhattacharyya, S. Adhikari, A. Banerjee, S. More, A. Kumar, **E. O. Nadler** *et al.* *The Signatures of Self-Interacting Dark Matter and Subhalo Disruption on Cluster Substructure*. 2022, [ApJ](#), 932, 30.

12. J. F. Wu, J. E. G. Peek, E. J. Tollerud, Y.-Y. Mao, **E. O. Nadler** et al. *Extending the SAGA Survey (xSAGA). I. Satellite Radial Profiles as a Function of Host Galaxy Properties*. 2022, [ApJ](#), **927**, 121.
11. D. Nguyen, D. Sarnaik, K. K. Boddy, **E. O. Nadler**, and V. Gluscevic. *Observational constraints on dark matter scattering with electrons*. 2021, [PRD](#), **104**, 103521.
- *10. A. Drlica-Wagner et al. (DELVE Collaboration, incl. **E. O. Nadler**). *The DECam Local Volume Exploration Survey: Overview and First Data Release*. 2021, [ApJS](#), **256**, 2.
9. Y. Wang, **E. O. Nadler**, Y.-Y. Mao, S. Adhikari, R. H. Wechsler et al. *UniverseMachine: Predicting Galaxy Star Formation over Seven Decades of Halo Mass with Zoom-in Simulations*. 2021, [ApJ](#), **915**, 116.
 - Major contributions: Interpreted dwarf galaxy star formation history predictions; analyzed simulations.
8. E. Darragh-Ford, **E. O. Nadler**, S. McLaughlin, and R. H. Wechsler. *Searching for Dwarfs in Gaia DR2 Phase-space Data using Wavelet Transforms*. 2021, [ApJ](#), **915**, 48.
 - Major contributions: Piloted and developed search algorithm; predicted number of detected dwarfs.
7. S. Das & **E. O. Nadler**. *Constraints on the epoch of dark matter formation from Milky Way satellites*. 2021, [PRD](#), **103**, 043517.
 - Major contributions: Derived and interpreted constraints on dark matter formation redshift.
6. K. Maamari, V. Gluscevic, K. K. Boddy, **E. O. Nadler**, and R. H. Wechsler. *Bounds on Velocity-dependent Dark Matter–Proton Scattering from Milky Way Satellite Abundance*. 2021, [ApJL](#), **907**, L46.
 - Major contributions: Developed numerical techniques to constrain interacting dark matter models.
- *5. Y.-Y. Mao, M. Geha, R. H. Wechsler, B. Weiner, E. J. Tollerud, **E. O. Nadler** et al. (SAGA Survey). *The SAGA Survey. II. Building a Statistical Sample of Satellite Systems around Milky Way–like Galaxies*. 2021, [ApJ](#), **907**, 85.
 - Major contributions: Provided theoretical predictions for SAGA satellite populations and interpreted results.
- *4. A. Drlica-Wagner, K. Bechtol, S. Mau, M. McNanna, **E. O. Nadler** et al. (DES Collaboration). *Milky Way Satellite Census. I. The Observational Selection Function for Milky Way Satellites in DES Y3 and Pan-STARRS DR1*. 2020, [ApJ](#), **893**, 47.
 - Major contributions: Developed machine-learning model of satellite detection sensitivity; analyzed simulations.
- *3. S. Mau et al. (DELVE Collaboration, incl. **E. O. Nadler**). *Two Ultra-Faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey*. 2020, [ApJ](#), **890**, 136.
2. C. E. Martínez-Vázquez et al. (DES Collaboration, incl. **E. O. Nadler**). *Search for RR Lyrae stars in DES ultrafaint systems: Grus I, Kim 2, Phoenix II, and Grus II*. 2019, [MNRAS](#), **490**, 2183.
1. K. M. Stringer et al. (DES Collaboration, incl. **E. O. Nadler**). *Identification of RR Lyrae Stars in Multi-band, Sparsely Sampled Data from the Dark Energy Survey Using Template Fitting and Random Forest Classification*. 2019, [AJ](#), **158**, 16.

White Papers

12. J. Han et al. *A Path to an All-Sky Survey with Roman*. 2026, [2602.21280](#).
11. J. Han et al. *NANCY: Next-generation All-sky Near-infrared Community survey*. 2023, [2306.11784](#).
10. A. Drlica-Wagner et al. *Report of the Topical Group on Cosmic Probes of Dark Matter for Snowmass 2021*. 2022, [2209.08215](#)
9. A. Banerjee et al. *Snowmass2021 Cosmic Frontier White Paper: Cosmological Simulations for Dark Matter Physics*. 2022, [2203.07049](#)
 - Major contributions: Developed simulation algorithm section and flowchart for tests of dark matter physics.
8. K. Bechtol et al. *Snowmass2021 Cosmic Frontier White Paper: Dark Matter Physics from Halo Measurements*. 2022, [2203.07354](#).

- Major contributions: Developed ultra-faint dwarf galaxy section and power spectrum visualization.
- 7. Y.-Y. Mao *et al.* *Snowmass2021: Vera C. Rubin Observatory as a Flagship Dark Matter Experiment*. 2022, [2203.07252](#).
- 6. K. Boddy *et al.* *Astrophysical and Cosmological Probes of Dark Matter*. 2022, [2203.06380](#).
- 5. S. Gezari *et al.* *R2-D2: Roman and Rubin – From Data to Discovery*. 2022, [2202.12311](#).
- 4. V. Gluscevic *et al.* *Cosmological Probes of Dark Matter Interactions: The Next Decade*. 2019, [1903.05140](#).
- 3. J. Simon *et al.* *Dynamical Masses for a Complete Census of Local Dwarf Galaxies*. 2019, [1903.04743](#).
- 2. K. Bechtol *et al.* *Dark Matter Science in the Era of LSST*. 2019, [1903.04425](#).
- 1. A. Drlica-Wagner *et al.* *Probing the Fundamental Nature of Dark Matter with the Large Synoptic Survey Telescope*. 2019, [1902.01055](#).
- Major contributions: Developed dwarf galaxy section and forecasted dark matter constraints.

Book Reviews

1. E. O. Nadler. *Dark Matter: Evidence, Theory, and Constraints* (Book Review). 2025, [Am. J. Phys 93, 763](#).

Interdisciplinary Publications

6. E. O. Nadler, D. Guilbeault, S. Ringold, T. R. Williamson *et al.* *Statistical or Embodied? Comparing Colorseeing, Colorblind, Painters, and Large Language Models in their Processing of Color Metaphors*. 2025, [Cognitive Science, 49, e70083](#).
5. D. Guilbeault, S. Delecourt, T. Hull, B. S. Desikan, M. Chu, and E. O. Nadler. *Online images amplify gender bias*. 2024, [Nature, 626, 1049](#).
4. E. O. Nadler, E. Darragh-Ford, B. S. Desikan *et al.* *Divergences in color perception between deep neural networks and humans*. 2023, [Cognition, 241, 105621](#).
3. M. Chu, B. S. Desikan, E. O. Nadler *et al.* *Signal in Noise: Exploring Meaning Encoded in Random Character Sequences with Character-Aware Language Models*. 2022, [Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 7120](#)
2. B. S. Desikan, T. Hull, E. O. Nadler *et al.* *comp-syn: Perceptually Grounded Word Embeddings with Color*. 2020, [Proceedings of the 28th International Conference on Computational Linguistics, 1744](#).
1. D. Guilbeault, E. O. Nadler *et al.* *Color associations in abstract semantic domains*. 2020, [Cognition, 201, 104306](#).